METRIC

MIL-I-51260A(EA)
8 February 1980
SUPERSEDING
MIL-I-51260(MU)
21 August 1967

MILITARY SPECIFICATION

INSECTICIDE, NALED (METRIC)

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one insecticide formulation of 85-percent naled (1,2-dibromo-2,2-dichloroethyl dimethyl phosphate).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-1337 - Drum, Shipping, Steel, With Polyethylene Insert

FSC 6840

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:55Z
Check the source to verify that this is the current version before use.

STANDARDS

FEDERAL

Fed. Test Method Std. No. 101 - Preservation, Packaging, and Packing Materials: Test Procedures

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads for 40" x 48" Pallets

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS (CFR)

40 CFR 162 - Regulations for the Enforcement of the Federal Insecticide, Fungicide, and Rodenticide Act
49 CFR 171 to 179 - Department of Transportation Hazardous Materials Regulations

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders for the above publications should cite "40 CFR 162" or "49 CFR 171 to 179.")

3. REQUIREMENTS

3.1 Composition. The insecticide formulation shall consist of naled and inert ingredients.

3.1.1 Naled. The insecticide formulation shall contain no less than 85.0 percent by weight naled when tested as specified in 4.2.4.1.

3.1.2 Inert ingredients. The inert ingredients shall consist of solvent (xylene), bromodichloroacetaldehyde, and trace amounts of chloral, methylated phosphoric polymers, dimethyl 2,2-dichlorovinyl phosphate, and other halogenated materials inherent in the manufacturing process.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:55Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the naled insecticide formulation produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For naled test (see 6.4). Sampling shall be conducted in accordance with table I. A representative specimen of approximately 100 milliliters (ml) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE I. Sampling for test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
Over 1,201	8

4.2.2.3 For container leakage test. Sampling shall be conducted in accordance with MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:55Z
Check the source to verify that this is the current version before use.

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled shipping container ready for shipment. Sample shipping containers shall be examined for the following defects using an AQL of 1.5 percent defective for defects grouped collectively:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Marking incorrect, missing, or illegible
(f) Labeling incorrect, missing, or illegible
(g) Palletization of level A drums not as specified

4.2.3.2 For naled test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.3.3 For container leakage test. The sample unit shall be one container. The sample containers selected in 4.2.2.3 shall be tested as specified in 4.2.5 using an AQL of 1.5 percent defective.

4.2.4 Naled test (see 6.4). The naled test shall be conducted as follows:

4.2.4.1 Assay.

(a) Gas chromatograph and column. Use a gas chromatograph having a thermal conductivity or flame ionization detector. An F & M Model 720 gas chromatograph or equal can be used by modifying an aluminum tubing column with a 6.35-millimeter (mm) "swagelok" tee secured to and over the injection side of the column. The cross arms of the "swagelok" should be bored through so that the column tubing will pass through the cross arms of the tee. Drill a 4.76-mm hole through the side of the column tubing visible through and at the junction of the branch arm (perpendicular to the cross arms) of the tee. Seal the open end of the branch arm with a septum sandwiched in the "swagelok" nut, using a back ferrule and a suitable washer to support the septum. Note: This column modification permits direct on-column injection on virtually all chromatographs. Care must be exercised when injecting through the septum. The void between the septum and the column tubing is not purged efficiently; any specimen expelled or released in this area will elute very slowly and cause tailing of the peaks. The needle should be inserted through the septum through the 4.76-mm hole in the column tubing and the shot should be expelled against the opposite wall of the column tubing for the shot to remain a compact slug. An F & M Model 820 gas chromatograph or equal can be used

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:55Z
Check the source to verify that this is the current version before use.

without modification for temperature-programmed or isothermal analysis. Note: Because the column cover of the F & M Model 720 gas chromatograph must be open to gain access to the injection septum, it is not possible to run column temperature isothermally.

(b) Chromatographic conditions. Recommended conditions for an F & M Model 720 gas chromatograph with column modified as described in (a) or for an F & M Model 820 gas chromatograph are shown in table II. Other equivalent gas chromatographs may be used but may require modification of conditions in order to obtain good peak shape, adequate resolution, and appropriate retention times. The column temperature may be programmed at 10°C per minute over the temperature range 65° to 225°C.

TABLE II. Chromatographic conditions

Characteristic	Condition
Column dimensions, millimeters	610 by 6.35 OD
Column material	Aluminum
Column support	HMDS treated "Chromsorb W", 60 to 80 mesh
Column coating	20 percent SE-52 silicone
Helium flow rate, ml per minute	100
Detector oven temperature	275°C
Injector temperature	220°C
Column temperature	190°C
Detector current, milliamperes	200
Attenuation setting	2
Chart speed, mm per minute	12.7

(c) Preparation of standards. Prepare four standard naled solutions using a standard naled sample of known purity (see 6.5) in the concentration range of 20 to 80 mg per milliliter (approximately 20, 40, 60, and 80 mg per milliliter) using acetone or benzene as the solvent.

(d) Preparation of specimen. Prepare a solution of the specimen in acetone or benzene so that the naled content falls in the range of 20 to 80 mg per milliliter.

(e) Procedure. Use the standard solutions prepared in (c) and the specimen solution prepared in (d). With the column stabilized and operating under the conditions described in table II, inject two 30-microliter aliquots of the 80 mg per milliliter standard to equilibrate the column. Perform the analysis by injecting 30-microliter aliquots in the following order: 80 mg per milliliter standard, specimen

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:55Z
Check the source to verify that this is the current version before use.

solution, 60 mg per milliliter standard, specimen solution, 40 mg per milliliter standard, specimen solution, and 20 mg per milliliter standard. Measure the areas of the four standard and three specimen peaks. Using the data from the four standards, prepare a calibration curve of concentration of naled in milligrams per milliliter versus peak area of injected standard. Use the calibration curve to calculate the percent by weight naled in the specimen as follows:

$$\text{Percent naled} = \frac{100A}{B}$$

where: A = Concentration of naled in milligrams per milliliter corresponding to the peak area of the specimen and
B = Concentration of the specimen in milligrams per milliliter in the specimen solution.

Notes: (1) A new column will not give reproducible results until it has been baked overnight at 250° to 270°C with a slow flow of helium and then pretreated with a large quantity of naled. The pretreatment is done by setting the column temperature at 180° to 200°C, injecting several 10 to 20 mg portions of naled over a period of 1/2 hour, and then allowing the baseline drawn by the recorder to come back to normal. (2) At the beginning of each set of injections, the first one or two results should be disregarded as they will not normally be typical. (3) If, at any time, the peak shapes are not typical, the results should be disregarded and the samples rerun.

4.2.5 Container leakage test. Test the containers for leakage in accordance with method 5009.1 of Fed. Test Method Std. No. 101 using the static technique.

5. PACKAGING

5.1 Packing. The insecticide shall be packed level A or C as specified (see 6.2).

5.1.1 Level A. The insecticide shall be packed level A in 18.9- or 56.8-liter quantities as specified (see 6.2).

5.1.1.1 Quantities of 18.9 liters. A quantity of 18.9 (+0.2 or -0) liters of insecticide shall be packed in nominal 18.9-liter capacity steel shipping drum with a polyethylene insert. The drum and insert shall conform to type II, class 1 of PPP-C-1337. The closures shall be secured to a torque as specified by the manufacturer of the drum. Each closure shall be protected by a pilfer-deterrent seal. There shall be no evidence of leakage when tested as specified in 4.2.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:55Z
Check the source to verify that this is the current version before use.

5.1.1.2 Quantities of 56.8 liters. A quantity of 56.8 (+0.5 or -0) liters of insecticide shall be packed in a nominal 56.8-liter capacity steel shipping drum with a polyethylene insert. The drum and insert shall conform to type II, class 2A of PPP-C-1337. The closure shall be secured to a torque as specified by the manufacturer of the drum. Each closure shall be protected by a pilfer-deterrent seal. There shall be no evidence of leakage when tested as specified in 4.2.5.

5.1.2 Level C. The insecticide shall be packed level C in 18.9- or 56.8-liter quantities as specified (see 6.2).

5.1.2.1 Quantities of 18.9 liters. A quantity of 18.9 (+0.2 or -0) liters of insecticide shall be packed as specified in 5.1.1.1 except that the drum shall conform to Department of Transportation (DOT) Specification 6D and shall have an inner liner conforming to or exceeding DOT Specification 2U.

5.1.2.2 Quantities of 56.8 liters. A quantity of 56.8 (+0.5 or -0) liters of insecticide shall be packed as specified in 5.1.1.2 except that the drum shall conform to DOT Specification 6D and shall have an inner liner conforming to DOT Specification 2SL.

5.2 Unitization. Drums of insecticide shall be palletized in accordance with the applicable requirements of MIL-STD-147.

5.3 Marking. Drums shall be marked in accordance with MIL-STD-129 and DOT regulations. Each drum shall be marked to show the date of manufacture of the insecticide and the lot or batch number.

5.3.1 Labeling. All containers shall be labeled in compliance with 40 CFR 162 and 49 CFR 171 to 179. Labels shall be approved and registered in accordance with 40 CFR 162. As a minimum, each container shall be labeled with the following information:

(a) The name and percent by weight of each active ingredient
(b) The name and percent by weight of each inactive ingredient
(c) National stock number, military specification number, contract number, and lot number
(d) Manufacturer's name and address
(e) EPA registration and establishment number
(f) Directions for use against house flies, gnats, midges, adult stable flies, and adult mosquitoes and application instructions for aircraft and ground ultra low volume and diluted formulations and ground thermal fog formulations.
(g) Precautionary information

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:55Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. Naled insecticide is intended for use against the insects listed in 5.3.1(f).

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Level of packing required (see 5.1)
(c) Unit quantity required (see 5.1.1 and 5.1.2)

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Sampling and testing precautions. Naled causes skin and eye damage. It may be fatal if swallowed or absorbed through the skin. Care must be exercised to avoid getting naled in eyes, on skin, or on clothing. Use waterproof gloves and face shield or goggles when handling naled. In case of eye contact, immediately flush eyes with plenty of water and see a physician. For skin contact, immediately remove contaminated clothing and flush affected area with plenty of water.

6.5 Standard sample of naled. A sample of naled of known purity for use in the assay test is available from Chevron Chemical Company, Ortho Division, 575 Market St., San Francisco, CA 94105.

6.6 Storage limitation. This material may be purchased for depot storage for periods up to 2 years. It is recommended that seasonal requirements be purchased on a year-to-year basis.

Custodian:

Army - EA

Review activities:

Army - MD
DLA - GS

Preparing activity:

Army - EA

Project No. 6840-A341

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:55Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-I-51260A(EA) Insecticide, Naled (Metric)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426

EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:55Z
Check the source to verify that this is the current version before use.